Linear Algebra and Its Applications

Section 1.3: Vector Equations

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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1.3 Vector equations

A vector is an ordered list; its coordinates are added and multiplied by scalars coordinate by coordinate. We display vectors as columns unless explicitly discussing points. A linear combination $c_1v_1+\cdots+c_kv_k$ uses scalar weights c_i. The span is the *set of all* combinations, not merely the finite list of generating vectors.

To test whether b is in a span, put the generators into the columns of a matrix and put b in the augmented column. Consistency of that system means the needed weights exist. An inconsistent system means b is outside the span. Neither the weights nor a solution should be guessed from a picture when exact coordinates are supplied.

Practice problem 1.3.P1: Commutativity from the coordinate definition

Take arbitrary $u = (u_1, \ldots, u_n)$ and $v = (v_1, \ldots, v_n)$. Their ith sum coordinate is u_i+v_i. Real-number commutativity gives $u_i + v_i = v_i + u_i$ for each i. Thus

$$u + v = (u_1 + v_1, \ldots, u_n + v_n) = (v_1 + u_1, \ldots, v_n + u_n) = v + u.$$

Conclusion. The vector identity holds because all corresponding coordinates are equal, not because the vector symbols can be rearranged without reason.

Practice problem 1.3.P2: Find the height that lies in a span

Seek weights x_1, x_2, x_3 and reduce

$$\left[\begin{array}{rrr|r} 1 & 5 & -3 & -4 \\ -1 & -4 & 1 & 3 \\ -2 & -7 & 0 & h \end{array}\right].$$

Use $R_2 \leftarrow R_2 + R_1$ and $R_3 \leftarrow R_3 + 2R_1$:

$$\left[\begin{array}{rrr|r} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 3 & -6 & h-8 \end{array}\right].$$

Now $R_3 \leftarrow R_3 - 3R_2$ gives $[0\ 0\ 0 \mid h - 5]$. Consequently consistency requires $h = 5$. If $h = 5$, choose $x_3 = 0$; row 2 gives $x_2 = -1$, and row 1 gives $x_1 = 1$. Indeed $v_1 - v_2 = (-4, 3, 5)^T$.

Conclusion. Exactly $h = 5$ works.

Exercise 1.3.1: Add vectors and subtract twice a vector

Add coordinates in the same positions, and first double v for the second expression:

$$u + v = \begin{bmatrix} -1 + (-3) \\ 2 + (-1) \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}, \qquad u - 2v = \begin{bmatrix} -1 - 2(-3) \\ 2 - 2(-1) \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}.$$

Conclusion. $u + v = (-4, 1)^T$ and $u - 2v = (5, 4)^T$.

Exercise 1.3.2: A second coordinate calculation

Coordinatewise addition and scalar multiplication give

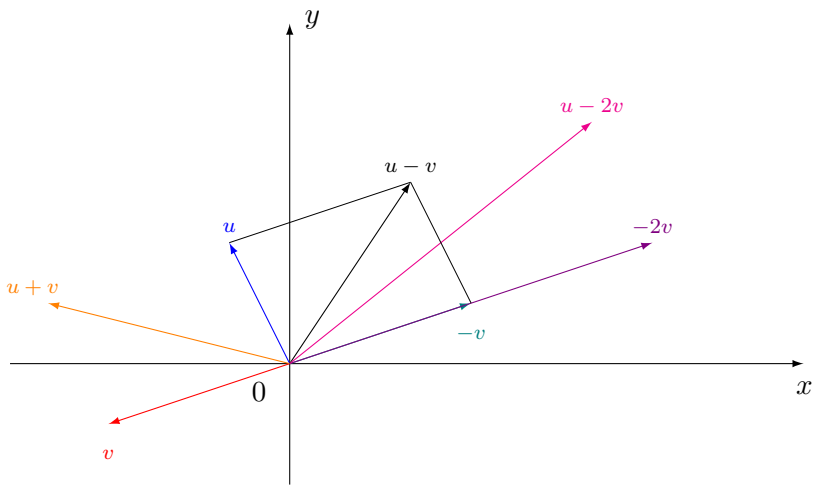
$$u+v=\begin{bmatrix}3+2\\2-1\end{bmatrix}=\begin{bmatrix}5\\1\end{bmatrix},\qquad u-2v=\begin{bmatrix}3-4\\2-2(-1)\end{bmatrix}=\begin{bmatrix}-1\\4\end{bmatrix}.$$

Conclusion. $u+v=(5,1)^T$ and $u-2v=(-1,4)^T$.

Exercise 1.3.3: Seven vectors and their parallelogram

Compute the coordinates before drawing. Every arrow below begins at the origin and ends at its vector coordinates.

vector	x	y
u	-1	2
v	-3	-1
$-v$	3	1
$-2v$	6	2
$u+v$	-4	1
$u-v$	2	3
$u-2v$	5	4

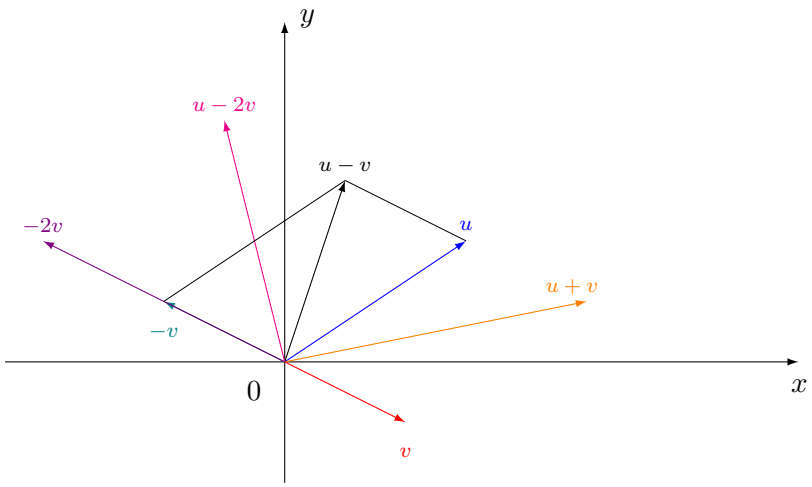


The thin extra sides join u to $u-v$ and $-v$ to $u-v$. They complete the parallelogram with vertices $0, u, u-v, -v$. The coordinate table fixes the exact endpoints even where arrows cross.

Exercise 1.3.4: A second arrow diagram

Compute the coordinates before drawing. Every arrow below begins at the origin and ends at its vector coordinates.

vector	x	y
u	3	2
v	2	-1
$-v$	-2	1
$-2v$	-4	2
$u+v$	5	1
$u-v$	1	3
$u-2v$	-1	4



The thin extra sides join u to $u-v$ and $-v$ to $u-v$. They complete the parallelogram with vertices $0, u, u-v, -v$. The coordinate table fixes the exact endpoints even where arrows cross.

Exercise 1.3.5: Read a vector equation coordinate by coordinate

The left side has coordinates $(3x_1+5x_2, -2x_1, 8x_1-9x_2)$. Two vectors are equal exactly when each corresponding coordinate is equal. Thus the equivalent system is

$$3x_1+5x_2=2, \qquad -2x_1=-3, \qquad 8x_1-9x_2=8.$$

Conclusion. These three scalar equations express precisely the given vector equation.

Exercise 1.3.6: Three weights but two coordinate equations

Multiply each column by its weight, add the two coordinates separately, and compare with the zero vector:

$$3x_1+7x_2-2x_3=0, \qquad -2x_1+3x_2+x_3=0.$$

Conclusion. The two scalar equations are equivalent to the vector equation; there are two equations even though there are three unknown weights.

Exercise 1.3.7: Read four vectors on an oblique grid

The grid consists of lines parallel to u and lines parallel to v. Following a line parallel to u changes only the u coefficient; following a line parallel to v changes only the v coefficient. Use the marked $u, v, -u, -v, -2v, 2v$ to fix unit steps. Counting those steps, including the halfway position of c, gives

$$a = u - 2v, \qquad b = 2u - 2v, \qquad c = 2u - \frac{7}{2}v, \qquad d = 3u - 4v.$$

For example, the line through a parallel to u meets the v-axis at $-2v$, and moving one u step from there reaches a. The vectors u, v point in distinct directions, so their two axes are not parallel. Through any plane point draw the corresponding two grid lines; they give one coefficient along each axis.

Conclusion. The displayed combinations describe a, b, c, d. Yes, every vector in $\mathbb{R}^2$ is a linear combination of these two nonparallel vectors.

Exercise 1.3.8: Read four more vectors on the grid

Count displacements on the same two oblique axes as in Exercise 7. The marked points give

$$w = -u + 2v, \qquad x = -2u + 2v, \qquad y = -2u + \frac{7}{2}v, \qquad z = -3u + 4v.$$

For instance, starting at $2v$ and moving one step in the $-u$ direction reaches w; moving one more step reaches x. The halfway placement of y in the v direction gives coefficient $7/2$, not 3 or 4. As the two generating directions are nonparallel, their grid extends across the whole plane.

Conclusion. The four combinations are as displayed, and the span is all of $\mathbb{R}^2$.

Exercise 1.3.9: Collect coefficients into three columns

Read down the coefficients of each variable, including its zero entry:

$$x_1 \begin{bmatrix} 0 \\ 4 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix} + x_3 \begin{bmatrix} 5 \\ -1 \\ -8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The first coordinate reproduces $x_2 + 5x_3 = 0$, the second reproduces $4x_1 + 6x_2 - x_3 = 0$, and the third reproduces the last equation. Thus no equation or coefficient has been omitted.

Exercise 1.3.10: Encode a nonhomogeneous system as a vector equation

Collect the coefficients down the three equations:

$$x_1 \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ -7 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ 5 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}.$$

Equality of each coordinate is exactly the corresponding original scalar equation, so the vector equation is equivalent to the system.

Exercise 1.3.11: A span test with a selected-answer discrepancy

What we must show. Decide whether weights exist, rather than whether b equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ -2 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_3. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2 - 5t_1 \\ x_2 = 3 - 4t_1 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

For a concrete combination take $x_3 = 0$, giving $b = 2a_1 + 3a_2$. Therefore b belongs to the span. The selected-answer entry for Exercise 11 in the supplied PDF says "No, but the explicit coordinate check $2(1, -2, 0)^T + 3(0, 1, 2)^T = (2, -1, 6)^T$ shows that entry is inconsistent with the printed data.

Exercise 1.3.12: A span test with unique weights

What we must show. Decide whether weights exist, rather than whether b equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 1 & -2 & 5 & 9 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 0 & 0 & 11 & -2 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 1 & \frac{7}{3} & -\frac{5}{3} \\ 0 & 0 & 11 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{11})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 1 & \frac{7}{3} & -\frac{5}{3} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-6)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & \frac{109}{11} \\ 0 & 1 & \frac{7}{3} & -\frac{5}{3} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (\frac{7}{3})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & \frac{109}{11} \\ 0 & 1 & 0 & -\frac{41}{33} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{245}{33} \\ 0 & 1 & 0 & -\frac{41}{33} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = \frac{245}{33} \\ x_2 = -\frac{41}{33} \\ x_3 = -\frac{2}{11} \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Thus $b = (245/33)a_1 - (41/33)a_2 - (2/11)a_3$, so it is a linear combination.

Exercise 1.3.13: An inconsistent column-span test

What we must show. Decide whether weights exist, rather than whether b equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ -2 & 8 & -4 & -3 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ 0 & 0 & 0 & 3 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 1 & \frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 3 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{3})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 1 & \frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 0 \\ 0 & 1 & \frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{7}{3})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 0 \\ 0 & 1 & \frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-4)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{26}{3} & 0 \\ 0 & 1 & \frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

There are no weights producing b, so it is not a linear combination of the columns.

Exercise 1.3.14: An explicit combination with one free weight

What we must show. Decide whether weights exist, rather than whether b equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rrr|r} 1 & 0 & 5 & 2 \\ -2 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_3. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2 - 5t_1 \\ x_2 = 3 - 4t_1 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

For a concrete combination take $x_3 = 0$, giving $b = 2a_1 + 3a_2$. Therefore b belongs to the span.

Exercise 1.3.15: Find the height in a generated plane

Write $x_1a_1 + x_2a_2 = b$ and reduce the augmented matrix:

$$\left[\begin{array}{rr|r} 1 & -5 & 3 \\ 3 & -8 & -5 \\ -1 & 2 & h \end{array}\right].$$

Use $R_2 \leftarrow R_2 - 3R_1$ and $R_3 \leftarrow R_3 + R_1$. The last two rows become $[0\ 7 \mid -14]$ and $[0\ -3 \mid h + 3]$. The former gives $x_2 = -2$, then the first row gives $x_1 = -7$. The last row requires $6 = h + 3$, or $h = 3$. Conversely these weights then give $-7a_1 - 2a_2 = (3, -5, 3)^T$ exactly.

Conclusion. $h = 3$, and only that height lies in the plane.

Exercise 1.3.16: Find a first coordinate in a generated plane

From the second coordinate of $x_1v_1 + x_2v_2 = y$, we immediately get $x_2 = -3$. The third coordinate is $-2x_1 + 7x_2 = -5$, so $-2x_1 - 21 = -5$ and $x_1 = -8$. The first coordinate then requires $h = x_1 - 2x_2 = -8 + 6 = -2$. Substitution verifies all three coordinates.

Conclusion. Exactly $h = -2$ works, with $y = -8v_1 - 3v_2$.

Exercise 1.3.17: Five combinations and their weights

Choose five different pairs of weights and compute their coordinates. The choice of weights itself proves membership in the span.

c_1	c_2	$c_1v_1 + c_2v_2$		
0	0	0	0	0
1	0	3	1	2
0	1	−4	0	1
1	1	−1	1	3
2	−1	10	2	3

Conclusion. These are five explicit vectors with their generating weights. A sketch is not needed or requested.

Exercise 1.3.18: Five combinations of a second generating pair

Choose five different pairs of weights and compute their coordinates. The choice of weights itself proves membership in the span.

c_1	c_2	$c_1v_1 + c_2v_2$		
0	0	0	0	0
1	0	1	1	−2
0	1	−2	3	0
1	1	−1	4	−2
2	−1	4	−1	−4

Conclusion. These are five explicit vectors with their generating weights. A sketch is not needed or requested.

Exercise 1.3.19: Dependent directions generate a line

We have $v_2 = (3/2)v_1$, since multiplying each coordinate of $(8, 2, -6)^T$ by $3/2$ gives $(12, 3, -9)^T$. Therefore $c_1v_1 + c_2v_2 = (c_1 + (3/2)c_2)v_1$. Every combination lies on the line through the origin in direction v_1, and every point on that line occurs by taking $c_2 = 0$.

Conclusion. The span is the line $\{t(8, 2, -6)^T : t \in \mathbb{R}\}$, not a plane.

Exercise 1.3.20: Two nonparallel vectors generate a plane

The vectors $(1, 1, -2)^T$ and $(-2, 3, 0)^T$ are not scalar multiples: a multiple of the first with third coordinate zero would have multiplier zero and hence be the zero vector. Two nonparallel directions

through the origin generate a plane. To identify it algebraically, observe that $n = (6, 4, 5)^T$ satisfies $n \cdot v_1 = 6 + 4 - 10 = 0$ and $n \cdot v_2 = -12 + 12 = 0$. Thus every combination obeys $6x + 4y + 5z = 0$. Conversely, given a point on this plane, choose $c_1 = -z/2$ and $c_2 = (y - c_1)/3$. Its second and third coordinates match, and the plane equation forces the first coordinate to match as well.

Conclusion. The span is the plane through the origin with equation $6x + 4y + 5z = 0$.

Exercise 1.3.21: Express every plane vector using two generators

Seek scalars a, b with $a(2, -1)^T + b(2, 1)^T = (h, k)^T$. The two equations are $2a + 2b = h$ and $-a + b = k$. The second gives $b = a + k$; substitute this into the first to get $4a + 2k = h$. Thus

$$a = \frac{h - 2k}{4}, \qquad b = \frac{h + 2k}{4}.$$

These real numbers are defined for every real h, k. Their substitution gives the desired coordinates exactly.

Conclusion. Every $(h, k)^T$ lies in the span, with the displayed weights.

Exercise 1.3.22: A nonzero-entry matrix with a proper span

Take

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

All entries of A are nonzero. Each linear combination of its columns has form $(t, t, t)^T$, where t is the sum of the three weights. But b does not have three equal coordinates. Therefore no combination can produce b.

Conclusion. The displayed A, b meet both requirements.

Exercise 1.3.23: Vector notation and span definitions

(a) False. A column vector has shape 2×1, whereas the displayed row has shape 1×2. Changing the orientation requires a transpose; the two matrix shapes are not interchangeable.

(b) False. If $(-5, 2)$ were a scalar multiple of $(-2, 5)$, the first coordinate would require multiplier $5/2$, but the second would then be $25/2$, not 2. Thus the two points do not lie on one line through zero.

(c) True. The expression $(1/2)v_1$ is $(1/2)v_1 + 0v_2$; a coefficient may be zero.

(d) True. Coordinatewise expansion of the vector equation gives exactly the equations encoded by $[a_1\ a_2\ a_3 \mid b]$.

(e) False. If $u = v \neq 0$ the span is a line, and if $u = v = 0$ it is just the origin. A plane requires two independent directions.

Exercise 1.3.24: What linear combinations do and do not require

(a) False. With $u = (1, 0)^T, v = (0, 1)^T$, the combination $u + v = (1, 1)^T$ lies on neither generating axis. The span contains combinations involving both directions.

(b) True. $\mathbb{R}^5$ is the set of ordered lists of five real coordinates.

(c) True. By definition, b is in the span exactly when suitable weights exist, which is the same as consistency of the augmented system.

(d) False. Adding $u - v$ to v yields $u - v + v = u$, not generally v. For example $u = 0$, $v \neq 0$ gives zero as the result.

(e) False. All zero weights are allowed and produce the zero vector. Consequently every span contains zero.

Exercise 1.3.25: A generating set is different from its span

(a) Distinguish the generating set from its span.

The three columns are $(1, 0, -2)^T$, $(0, 3, 6)^T$, and $(-4, -2, 3)^T$. None equals $b = (4, 1, -4)^T$. Thus b is not in the three-element set $\{a_1, a_2, a_3\}$, which contains exactly three distinct vectors.

(b) Produce actual weights.

Direct calculation gives

$$-4a_1 - a_2 - 2a_3 = (-4, 0, 8)^T + (0, -3, -6)^T + (8, 4, -6)^T = (4, 1, -4)^T = b.$$

So $b \in W$. Also W contains all multiples of the nonzero vector a_1; different real multiples give different vectors, so W is infinite.

(c) Use the definition without reducing a matrix.

We have $a_1 = 1a_1 + 0a_2 + 0a_3$, which explicitly represents a_1 as a combination of the generators.

Conclusion. The finite generating set does not contain b, but its infinite span does contain b and all three generating columns.

Exercise 1.3.26: A target outside the span but a column inside

(a) Check the three coordinate equations for compatible weights.

Writing the weights as c_1, c_2, c_3, the first coordinate gives $2c_1 + 6c_3 = 10$, or $c_1 = 5 - 3c_3$. The second is $-c_1 + 8c_2 + 5c_3 = 3$, so after substitution $c_2 + c_3 = 1$. The third would require

$$c_1 - 2c_2 + c_3 = 7.$$

Its left side, using the first two equations, is instead $(5 - 3c_3) - 2(1 - c_3) + c_3 = 3$, independent of c_3. Thus it cannot equal 7; no weights exist, and $b \notin W$.

(b) Exhibit the trivial combination for a generator.

The second column is $0a_1 + 1a_2 + 0a_3$, so it is certainly in W.

Conclusion. b is outside the column span, although each generating column itself is automatically inside it.

Exercise 1.3.27: Translate mine outputs into a vector equation

(a) Interpret the scale factor together with the units.

Five days of mine 1 produce $5v_1 = (150, 3000)^T$: 150 metric tons of copper and 3000 kilograms of silver.

(b) Express the target as the sum of the two outputs.

The vector equation is

$$x_1 \begin{bmatrix} 30 \\ 600 \end{bmatrix} + x_2 \begin{bmatrix} 40 \\ 380 \end{bmatrix} = \begin{bmatrix} 240 \\ 2824 \end{bmatrix}.$$

The unknowns count days, so physically they must be nonnegative.

(c) Solve and check.

Divide the copper equation by 10: $3x_1 + 4x_2 = 24$. Multiply it by 200 and subtract the silver equation to get $420x_2 = 4800 - 2824 = 1976$. Therefore

$$x_2 = \frac{494}{105}, \qquad x_1 = \frac{24 - 4(494/105)}{3} = \frac{544}{315}.$$

Both are positive. In the copper equation their contributions sum to $30(544/315) + 40(494/105) = 240$; in the silver equation they sum to $600(544/315) + 380(494/105) = 2824$.

Conclusion. Mine 1 runs $544/315 \approx 1.727$ days and mine 2 runs $494/105 \approx 4.705$ days.

Exercise 1.3.28: Check three coal-plant output coordinates

(a) Track just the heat coordinate.

Burning x_1 tons of A and x_2 tons of B gives $27.6x_1 + 30.2x_2$ million Btu of heat.

(b) Keep the three output types in a fixed order.

With coordinates heat (million Btu), sulfur dioxide (g), and particulates (g), the output vector is

$$x_1 \begin{bmatrix} 27.6 \\ 3100 \\ 250 \end{bmatrix} + x_2 \begin{bmatrix} 30.2 \\ 6400 \\ 360 \end{bmatrix}.$$

(c) Set this vector equal to the observed output.

Use right side $(162, 23610, 1623)^T$. The heat and particulate equations, clearing the heat decimal, are

$$276x_1 + 302x_2 = 1620, \qquad 250x_1 + 360x_2 = 1623.$$

Subtracting 250 times the first from 276 times the second gives $23860x_2 = 42948$, so $x_2 = 9/5 = 1.8$. Substitution in either equation gives $x_1 = 39/10 = 3.9$. Now check the remaining equation rather than ignoring the extra datum:

$$3100(3.9) + 6400(1.8) = 12090 + 11520 = 23610.$$

The other outputs are $107.64 + 54.36 = 162$ and $975 + 648 = 1623$.

Conclusion. The plant burned 3.9 tons of anthracite and 1.8 tons of bituminous coal, satisfying all three measurements.

Exercise 1.3.29: Compute a mass-weighted average

The total mass is $4 + 2 + 3 + 5 = 14$ grams. Form the weighted sum one coordinate at a time:

$$\begin{aligned} 4v_1 + 2v_2 + 3v_3 + 5v_4 &= (8, -8, 16) + (-8, 4, 6) \\ &\quad + (12, 0, -6) + (5, -30, 0) \\ &= (17, -34, 16). \end{aligned}$$

Dividing by the total mass gives

$$\overline{v} = \left(\frac{17}{14}, -\frac{17}{7}, \frac{8}{7}\right).$$

The division is by total *mass*, not by the number of points.

Conclusion. The center of mass is $(17/14, -17/7, 8/7)$.

Exercise 1.3.30: A center of mass is a linear combination

Assume the total mass m is nonzero, as required to define the center of mass. Rewrite its formula as

$$\overline{v} = \frac{m_1}{m}v_1 + \cdots + \frac{m_k}{m}v_k.$$

The coefficients m_j/m are real scalars. This is precisely a linear combination of the given points viewed as vectors, so it belongs to their span. For positive physical masses the coefficients also sum to one and are nonnegative, although those extra facts are not needed for span membership.

Conclusion. Yes; the displayed normalized masses are the required weights.

Exercise 1.3.31: Move the center of a triangular plate

(a) Average the three equal point masses.

The uniform triangular plate has the same center as three 1-gram masses at the vertices, so

$$\overline{v} = \frac{(0, 1) + (8, 1) + (2, 4)}{3} = \left(\frac{10}{3}, 2\right).$$

(b) Include the plate's existing mass in the new balance.

The total mass after adding 6 grams will be 9 grams. The plate already has first moment $3\overline{v} = (10, 6)$. Let w_i be the added grams at vertex v_i. The new center $(2, 2)$ requires total first moment $(18, 18)$, so

$$w_1 + w_2 + w_3 = 6, \qquad 8w_2 + 2w_3 = 18 - 10 = 8,$$

$$w_1 + w_2 + 4w_3 = 18 - 6 = 12.$$

Subtract the mass equation from the last equation to obtain $3w_3 = 6$, so $w_3 = 2$. The horizontal equation then gives $8w_2 = 4$ and $w_2 = 1/2$. Finally $w_1 = 6 - 2 - 1/2 = 7/2$. All added masses are nonnegative, as the physical request requires.

Conclusion. Add 3.5 g at $(0, 1)$, 0.5 g at $(8, 1)$, and 2 g at $(2, 4)$. Their contribution is $(8, 12)$, making the final first moment $(18, 18)$ and the desired center $(2, 2)$.

Exercise 1.3.32: A plane-spanning diagram with nonunique weights

The figure shows v_1, v_2 pointing in nonparallel directions. Their span is therefore the whole plane, so in particular there are weights a, b with $av_1 + bv_2 = b_{\text{target}}$. Taking the third weight to be zero already gives a solution. The same plane contains v_3, so write $v_3 = pv_1 + qv_2$. For every $t \in \mathbb{R}$,

$$(a - pt)v_1 + (b - qt)v_2 + tv_3 = av_1 + bv_2 = b_{\text{target}}.$$

These are distinct coefficient triples because their third entries are distinct. Negative weights are allowed; the target need not lie between the positive arrows.

Conclusion. A solution exists, and it is not unique: there are infinitely many coefficient triples.

Exercise 1.3.33: Associativity and distributivity coordinatewise

(a) Associativity.

For each coordinate i, real-number associativity gives $(u_i + v_i) + w_i = u_i + (v_i + w_i)$. The ith coordinates of $(u + v) + w$ and $u + (v + w)$ are therefore equal for every i, which proves equality of the vectors.

(b) Distributivity over vector addition.

For each i, $c(u_i + v_i) = cu_i + cv_i$ by real-number distributivity. The left side is the ith coordinate of $c(u + v)$; the right side is the ith coordinate of $cu + cv$. Equality coordinate by coordinate gives $c(u + v) = cu + cv$.

Conclusion. Both vector laws follow from the corresponding real-number laws.

Exercise 1.3.34: Additive inverses and scalar associativity

(a) The additive inverse.

The ith coordinate of $u + (-u)$ is $u_i + (-u_i) = 0$. Similarly the ith coordinate of $(-u) + u$ is $-u_i + u_i = 0$. Both sums are therefore the vector $(0, \ldots, 0)$.

(b) Two successive scalar multiplications.

The ith coordinate of $c(du)$ is $c(du_i)$. Real multiplication is associative, so $c(du_i) = (cd)u_i$, which is the ith coordinate of $(cd)u$. This holds for every coordinate and for all real c, d, including zero and negative values.

Conclusion. $u + (-u) = (-u) + u = 0$ and $c(du) = (cd)u$.